Question 2-1

Write answers on this side of the paper only.

Do not write on either margin

For this problem, we will use the Bipartite Matching algorithm covered during lecture.

First, ~~the~~ I create a Bipartite Graph with 2 disjoint and independent sets of ~~n^2 vertices~~ the undeleted squares. We do so by iterating through all n^2 tiles on the checkboard and check if deleted[i,j] is true for tile i,j. If it is TRUE, we add ~~the~~ [i,j] into the set. We replicate this for the other set in the Bipartite Graph. This will take $O(n^2)$ time.

Then, we create an edge between 2 vertices in different sets if the 2 vertices are adjacent, which is only ~~these~~ in 4 cases, ~~if~~ given vertex [i,j] and vertex [k,l]. ~~i~~

Add an edge if $\pm(i-k) = 1$ and $\pm(j-l) = 1$

This ~~will take~~ $O(n^4)$ time as we will have to iterate through both sets of ~~$O(n^2)$~~ $O(n^2)$ vertices. ~~to see the~~ if

Next, we check for the number of undeleted tiles by iterating through ~~all the~~ the deleted[i,j] for all tiles [i,j]. This will take $O(n^2)$ time.

Lastly, we will use the algorithm covered in lecture of Bipartite Matching to compute the maximum matching of the graph. If the maximum matching is equal to the number of undeleted tiles in the checkerboard, we output TRUE. Else, we output FALSE. The algorithm takes $O(VE)$ $= O(n^2 \times 4n^2) = O(n^4)$ time.

Hence, the overall time complexity is $O(n^4)$.

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